DS	**Session**	Hashing	**Question Information**	Level 1 Challenge 100

Problem Description

Mathison has uncovered an ancient piece of paper on which are written N integers. This set of numbers will be referred to as A. Mathison learned in History class that a group of integers is unique if and only if their total is divisible by a mythological constant M.

Mathison tries to find out how many **distinct** triplets of numbers, from the piece of paper, have their sum divisible by *M*. Unfortunately, this problem is quite hard to crack and he needs your help.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 10^4$
- $0 \leq A[i] \leq 2 \times 10^9$

Input
The first line of the input file contains two space-separated integers, *N* and *M*, representing the number of integers and the mythical constant.
The next line contains *N* space-separated integers, where the ith integer represents $A[i]$.

Output
The output file should contain only one integer, the answer to Mathison's question.

Explanation for Test case 1

There are 26 *special* trios: (0, 1, 2); (0, 1, 8); (0, 2, 5); (0, 2, 6); (0, 2, 9); (0, 3, 7); (0, 5, 8); (0, 6, 8); (0, 8, 9); (1, 2, 7); (1, 3, 4); (1, 5, 6); (1, 5, 9); (1, 6, 9); (1, 7, 8); (2, 4, 8); (2, 5, 7); (2, 6, 7); (2, 7, 9); (3, 4, 5); (3, 4, 6); (3, 4, 9); (5, 6, 9); (5, 7, 8); (6, 7, 8); (7, 8, 9);

Note: Here we only show the **positions** (0-indexed) in each trio.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
10 5
1 10 4 3 2 5 0 1 9 5
```

EXPECTED OUTPUT

```
26
```

Test Case 2

INPUT (STDIN)

```
10 5
11 10 14 31 21 15 10 11 9 51
```

EXPECTED OUTPUT

```
31
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(i<k)
```

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
10	450	55

Code

C mathison_divisible_triplets.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXM 10005

long long cnt[MAXM];

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    for (int i = 0; i < n; i++) {
        long long val;
        scanf("%lld", &val);
        cnt[((val % m) + m) % m]++;
    }

    long long ans = 0;

    for (int i = 0; i < m; i++) {
        if ((3LL * i) % m == 0 && cnt[i] >= 3) {
            ans += cnt[i] * (cnt[i] - 1) * (cnt[i] - 2) / 6;
        }
    }

    for (int i = 0; i < m; i++) {
        int r3 = (m - (2 * i) % m) % m;
        if (r3 != i && cnt[i] >= 2 && cnt[r3] >= 1) {
            ans += (cnt[i] * (cnt[i] - 1) / 2) * cnt[r3];
        }
    }

    for (int i = 0; i < m; i++) {
        for (int j = i + 1; j < m; j++) {
            int k = (m - (i + j) % m) % m;
            int ick = (j < k);
            while( ick) {
                ans += cnt[i] * cnt[j] * cnt[k];
                break;
            }
        }
    }

    printf("%lld\n", ans);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc mathison_divisible_triplets.c -o mathison_divisible_triplets
PS C:\C Programming> .\mathison_divisible_triplets
10 5
11 18 14 31 21 15 10 11 9 51
26
PS C:\C Programming>